

Vieta's Formulas

16. The roots of $x^2 - x - 1 = 0$ are ϕ and $\hat{\phi}$ (the golden ratio and its conjugate). Compute $\phi^6 + \hat{\phi}^6$.

Let the two roots of

$$x^2 - x - 1 = 0$$

be ϕ and $\hat{\phi}$. We want to compute

$$\phi^6 + \hat{\phi}^6.$$

Since both roots satisfy the equation,

$$x^2 - x - 1 = 0,$$

we have

$$x^2 = x + 1.$$

Now repeatedly use this relation:

$$x^3 = x(x^2) = x(x + 1) = x^2 + x = (x + 1) + x = 2x + 1,$$

$$x^4 = x(2x + 1) = 2x^2 + x = 2(x + 1) + x = 3x + 2,$$

$$x^5 = x(3x + 2) = 3x^2 + 2x = 3(x + 1) + 2x = 5x + 3,$$

$$x^6 = x(5x + 3) = 5x^2 + 3x = 5(x + 1) + 3x = 8x + 5.$$

Therefore,

$$\phi^6 = 8\phi + 5$$

and

$$\hat{\phi}^6 = 8\hat{\phi} + 5.$$

Adding them gives

$$\phi^6 + \hat{\phi}^6 = 8(\phi + \hat{\phi}) + 10.$$

By Vieta's formulas, the sum of the roots of $x^2 - x - 1 = 0$ is

$$\phi + \hat{\phi} = 1.$$

Thus,

$$\phi^6 + \hat{\phi}^6 = 8(1) + 10 = \boxed{18}.$$

17. The roots of $x^2 - 4x + 2 = 0$ are r and s . Find a recurrence relationship for $r^n + s^n$ and use it to compute $r^{10} + s^{10}$.

Let

$$a_n = r^n + s^n,$$

where r and s are the roots of

$$x^2 - 4x + 2 = 0.$$

Since each root satisfies the equation,

$$x^2 = 4x - 2.$$

Multiplying both sides by x^n , we get

$$x^{n+2} = 4x^{n+1} - 2x^n.$$

This is true for both r and s , so

$$r^{n+2} = 4r^{n+1} - 2r^n$$

and

$$s^{n+2} = 4s^{n+1} - 2s^n.$$

Adding these equations gives

$$r^{n+2} + s^{n+2} = 4(r^{n+1} + s^{n+1}) - 2(r^n + s^n).$$

Therefore, the recurrence relationship is

$$\boxed{a_{n+2} = 4a_{n+1} - 2a_n}.$$

The initial values are

$$a_0 = r^0 + s^0 = 2$$

and, by Vieta's formulas,

$$a_1 = r + s = 4.$$

Now apply the recurrence:

$$a_2 = 4a_1 - 2a_0 = 4(4) - 2(2) = 12,$$

$$a_3 = 4a_2 - 2a_1 = 4(12) - 2(4) = 40,$$

$$a_4 = 4a_3 - 2a_2 = 4(40) - 2(12) = 136,$$

$$a_5 = 4a_4 - 2a_3 = 4(136) - 2(40) = 464,$$

$$a_6 = 4a_5 - 2a_4 = 4(464) - 2(136) = 1584,$$

$$a_7 = 4a_6 - 2a_5 = 4(1584) - 2(464) = 5408,$$

$$a_8 = 4a_7 - 2a_6 = 4(5408) - 2(1584) = 18464,$$

$$a_9 = 4a_8 - 2a_7 = 4(18464) - 2(5408) = 63040,$$

$$a_{10} = 4a_9 - 2a_8 = 4(63040) - 2(18464).$$

Thus,

$$a_{10} = 252160 - 36928 = \boxed{215232}.$$

Therefore,

$$\boxed{r^{10} + s^{10} = 215232}.$$

21. A cubic $x^3 + px^2 + qx + r = 0$ has roots 1, 2, 3. Find p , q , and r .

The cubic equation is

$$x^3 + px^2 + qx + r = 0,$$

and its roots are 1, 2, and 3.

Therefore, the polynomial can be written as

$$(x - 1)(x - 2)(x - 3) = 0.$$

First, multiply the first two factors:

$$(x - 1)(x - 2) = x^2 - 3x + 2.$$

Now multiply by $(x - 3)$:

$$(x^2 - 3x + 2)(x - 3).$$

Expanding,

$$\begin{aligned} x^3 - 3x^2 - 3x^2 + 9x + 2x - 6 \\ = x^3 - 6x^2 + 11x - 6. \end{aligned}$$

Comparing

$$x^3 - 6x^2 + 11x - 6$$

with

$$x^3 + px^2 + qx + r,$$

we get

$$\boxed{p = -6, \quad q = 11, \quad r = -6}.$$

24. The roots of $x^3 - 3x^2 + 7x - 1 = 0$ are r, s, t . Without factoring, compute $r^3 + s^3 + t^3$.

Let

$$a_n = r^n + s^n + t^n,$$

where r, s, t are the roots of

$$x^3 - 3x^2 + 7x - 1 = 0.$$

We are asked to find $a_3 = r^3 + s^3 + t^3$ **without factoring** the polynomial.

Since each root satisfies

$$x^3 - 3x^2 + 7x - 1 = 0,$$

we have

$$x^3 = 3x^2 - 7x + 1.$$

Applying this to each root,

$$r^3 = 3r^2 - 7r + 1,$$

$$s^3 = 3s^2 - 7s + 1,$$

$$t^3 = 3t^2 - 7t + 1.$$

Adding the three equations,

$$r^3 + s^3 + t^3 = 3(r^2 + s^2 + t^2) - 7(r + s + t) + 3.$$

Now compute the needed sums using Vieta's formulas.

From

$$x^3 - 3x^2 + 7x - 1 = 0,$$

we have

$$r + s + t = 3,$$

$$rs + rt + st = 7.$$

Next,

$$r^2 + s^2 + t^2 = (r + s + t)^2 - 2(rs + rt + st) = 3^2 - 2(7) = 9 - 14 = -5.$$

Therefore,

$$r^3 + s^3 + t^3 = 3(-5) - 7(3) + 3 = -15 - 21 + 3 = -33.$$

40. Let r_1, r_2, r_3, r_4 be the roots of $x^4 - 8x^3 + 14x^2 + 8x - 15 = 0$. Compute $r_1^2 + r_2^2 + r_3^2 + r_4^2$.

Let

$$a_n = r_1^n + r_2^n + r_3^n + r_4^n,$$

where r_1, r_2, r_3, r_4 are the roots of

$$x^4 - 8x^3 + 14x^2 + 8x - 15 = 0.$$

We are asked to find

$$a_2 = r_1^2 + r_2^2 + r_3^2 + r_4^2.$$

Using Vieta's formulas for

$$x^4 - 8x^3 + 14x^2 + 8x - 15 = 0,$$

we have

$$r_1 + r_2 + r_3 + r_4 = 8,$$

and

$$r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4 = 14.$$

Now use the identity

$$(r_1 + r_2 + r_3 + r_4)^2 = r_1^2 + r_2^2 + r_3^2 + r_4^2 + 2 \sum_{i < j} r_i r_j.$$

Therefore,

$$a_2 = (r_1 + r_2 + r_3 + r_4)^2 - 2(r_1r_2 + r_1r_3 + r_1r_4 + r_2r_3 + r_2r_4 + r_3r_4).$$

Substitute the values from Vieta's formulas:

$$a_2 = 8^2 - 2(14) = 64 - 28 = 36.$$

Hence,

$$\boxed{r_1^2 + r_2^2 + r_3^2 + r_4^2 = 36.}$$

Shortcut (worth remembering)

For any polynomial with roots $r_1, \dots, r_n$,

$$\boxed{\sum r_i^2 = (\sum r_i)^2 - 2 \sum_{i < j} r_i r_j.}$$

This identity, together with **Vieta's formulas**, is the fastest way to solve this type of problem.

41. Let r_1, r_2, r_3, r_4 be the roots of $x^4 - 6x^2 + 1 = 0$. Compute $r_1^4 + r_2^4 + r_3^4 + r_4^4$.

Let

$$S = r_1^4 + r_2^4 + r_3^4 + r_4^4,$$

where r_1, r_2, r_3, r_4 are the roots of

$$x^4 - 6x^2 + 1 = 0.$$

Since each root r_i satisfies the equation,

$$r_i^4 - 6r_i^2 + 1 = 0,$$

so

$$r_i^4 = 6r_i^2 - 1.$$

Adding this equation for all four roots gives

$$r_1^4 + r_2^4 + r_3^4 + r_4^4 = 6(r_1^2 + r_2^2 + r_3^2 + r_4^2) - 4.$$

Now use Vieta's formulas. For

$$x^4 + 0x^3 - 6x^2 + 0x + 1 = 0,$$

we have

$$r_1 + r_2 + r_3 + r_4 = 0$$

and

$$\sum_{i < j} r_i r_j = -6.$$

Using

$$r_1^2 + r_2^2 + r_3^2 + r_4^2 = (r_1 + r_2 + r_3 + r_4)^2 - 2 \sum_{i < j} r_i r_j,$$

we get

$$r_1^2 + r_2^2 + r_3^2 + r_4^2 = 0^2 - 2(-6) = 12.$$

Therefore,

$$S = 6(12) - 4 = 72 - 4 = 68.$$

Hence,

$$\boxed{r_1^4 + r_2^4 + r_3^4 + r_4^4 = 68}.$$